



Identification of the MYB gene family in *Sorghum bicolor* and functional analysis of *SbMYBAS1* in response to salt stress

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Abstract

Salt stress adversely affects plant growth and development. It is necessary to understand the underlying salt response mechanism to improve salt tolerance in plants. MYB transcription factors can regulate plant responses to salt stress. However, only a few studies have explored the role of MYB TFs in *Sorghum bicolor* (L.) Moench. So we decided to make a systematic analysis and research on the sorghum MYB family. A total of 210 MYB genes in sorghum were identified in this study. Furthermore, 210 MYB genes were distributed across ten chromosomes, named *SbMYB1*-*SbMYB210*. To study the phylogeny of the identified TFs, 210 MYB genes were divided into six subfamilies. We further demonstrated that *SbMYB* genes have evolved under strong purifying selection. *SbMYBAS1* (*SbMYB119*) was chosen as the study object, which the expression decreased under salt stress conditions. Further study of the *SbMYBAS1* showed that *SbMYBAS1* is located in the nucleus. Under salt stress conditions, Arabidopsis plants overexpressed *SbMYBAS1* showed significantly lower dry/fresh weight and chlorophyll content but significantly higher membrane permeability, MDA content, and Na⁺/K⁺ ratio than the wild-type Arabidopsis plants. Yeast two-hybrid screening result showed that *SbMYBAS1* might interact with proteins encoded by SORBI_302G184600, SORBI_3009G247900 and SORBI_3004G59600. Results also showed that *SbMYBAS1* could regulate the expression of *AtGSTU17*, *AtGSTU16*, *AtP5CS2*, *AtUGT88A1*, *AtUGT85A2*, *AtOPR2* and *AtPCR2* under salt stress conditions. This work laid a foundation for the study of the response mechanism of sorghum MYB gene family to salt stress.

Key message

This report shows detailed characterization of sorghum MYB gene family and demonstrates that *SbMYBAS1* plays an important role in negative regulation of salt tolerance by reducing ROS scavenging ability and K⁺ content in Arabidopsis.

Keywords Sorghum · Transcription factor · MYB gene family · Salt stress tolerance

Abbreviations

MYB Myeloblastosis
 TF Transcription factor

HMM	Hidden Markov Model
ML	Maximum likelihood
GSDS	Gene structure display server
MEME	Multiple Em for Motif Elicitation
CDS	Coding sequence
MW	Molecular weight
pI	Isoelectric point
MCSscanX	Multiple collinearity scan toolkit
FPKM	Fragments Per Kilobase of exon model per Million mapped fragments
gDNA	Genomic DNA
RNA	Ribonucleic acid
qRT-PCR	Real-time quantitative reverse transcription PCR
ABRE	Abscisic acid responsive element
MeJARE	Methyl jasmonate responsive element

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DSRE	Defense and stress responsive element
AURE	Auxin responsive element
SARE	Salicylic acid responsive element
GIRE	Gibberellin responsive element

Introduction

A continuous rise in the global population and deterioration of the natural environment has resulted in a global crisis for food and energy. Sorghum is an energy and food crop that has gained wide attention due to its drought, waterlogging, and saline-alkali resistance (Huang 2018). Sorghum, the fifth largest cereal crop globally, is a staple food for people in arid and semi-arid regions. Besides being an important cash crop, sorghum is also widely used for feed, forage, and fuel (Hao et al. 2021).

Sorghum is an important plant for researchers due to its lower production cost and higher energy gain. The primary factors affecting the yield of sorghum are biological (pests and diseases) and abiotic (drought, high temperature, high salt, and so on) stresses. First reported in 2013, sugarcane aphid (*Melanaphis sacchari*) remains to be the greatest threat to sorghum yield in all high-yielding regions of the United States (Kiani and Szczepaniec 2018). Drought and high temperature are the most important abiotic stresses limiting sorghum yield in arid and semi-arid regions (Prasad et al. 2021). However, the high-yielding region of sorghum worldwide has low or medium-salt content. Thus, salt stress has an important impact on sorghum yield (Amombo et al. 2022). In sorghum, high salt levels affect biomass, plant height, and leaf area (Sun et al. 2019). In addition, high salt content limits the self-healing ability of sorghum seedlings (Zhao et al. 2014), reducing its pigment content and photosynthetic efficiency (Stefanov et al. 2021). High salt also inhibits the alpha-amylase activity of the sorghum plants, reducing their starch yield (Punia et al. 2021).

In adapt to salt stress, plants have evolved multiple mechanisms to reduce the damage caused due to salt stress. For instance, plants reduce damage in high-salt environments by regulating ion and osmotic homeostasis, activating stress resistance mechanisms mediated by plant hormones (ABA signaling and BR signaling), triggering dynamic changes in the cytoskeleton, regulating cell walls, and so on (Zhao et al. 2021). These responses by plants are controlled by genes encoded by transcriptional activators or repressors. In addition, salt stress induces a series of downstream changes by activating or repressing the expression of transcription factors (TFs). WRKY (Baillo et al. 2020), GRAS (Fan et al. 2021a), NAC (Sanjari et al. 2019), bHLH (Fan et al. 2021b), bZIP (Wang et al. 2011), and MYB (Singh et al. 2020) are the TFs that are primarily associated with the abiotic stress in sorghum.

The MYB family TFs, one of the most prominent members of the plant TF family, affect plant growth and development. The first MYB TF cloned in plants was *ZmMYB1* in maize (Paz-Ares et al. 1987). MYB domain consists of 1–4 repeats of approximately 52 amino acids, forming three α -helices, and the second and third helix forms a helix-turn-helix (HTH) (Ogata et al. 1996). According to the adjacent repeats, the MYB family can be divided into four subfamilies: 1R-MYB (MYB-related protein), 2R-MYB (R2R3-MYB), 3R-MYB, 4R-MYB, and the most abundant 2R-MYB in plants (Lipsick 1996; Martin and Paz-Ares 1997).

MYB TFs in abiotic stress were involved in the following functions: regulation of flavonoids, cuticle synthesis, drought stress (Nakabayashi et al. 2014; Cui et al. 2016), salt tolerance via ABA signaling pathway regulation (Cui et al. 2013), cold tolerance via CBF expression (Agarwal et al. 2006), and participation in the heat stress response via calcium signaling regulation (Liao et al. 2017). In sorghum, MYB TFs primarily include *SbMYB60*, which regulates the composition of plant cell secondary walls by affecting the phenylpropane synthesis (Scully et al. 2016). The MYB TF encoded by the yellow seed 1 (*y1*) gene in sorghum can enhance maize resistance to leaf blight by regulating anthocyanin synthesis (Ibraheem et al. 2015). The poplar gene *PagMYB151* improved the salt tolerance of poplar trees by promoting the accumulation of proline content (Hu et al. 2023). Obtaining a new 1R-MYB TF gene from *Fragaria vesca* also enhanced salt adaptability and tolerance in Arabidopsis (Li et al. 2023). The R2R3-MYB gene *SiMYB16* in foxtail millet enhances salt tolerance by regulating pathways such as phenylpropane biosynthesis, fatty acid extension, and phenylalanine metabolism (Yu et al. 2023). Overexpression of peanut MYB related transcription factor *AhMYB30* in Arabidopsis enhanced plant tolerance to cold and salt stress (Chen et al. 2023). Sorghum MYB gene *SbMYBHv33* negatively regulated the salt tolerance of sorghum and Arabidopsis by regulating the accumulation of reactive oxygen and ions (Zheng et al. 2023). The MYB gene family has a variety of functions, which may enhance or decrease the salt tolerance of plants by regulating various metabolism ways.

Although the MYB TF plays a crucial role in abiotic stress, few studies have explored the MYB TF in sorghum under salt stress conditions. In the current study, 210 genes from the sorghum MYB family were identified and categorized into six groups. The conserved motifs and structures of the genes within the same group were found to be identical, indicating that these genes were evolutionarily conserved. The transcriptome data of sorghum under salt stress conditions revealed the expression level of the *SbMYBAS1* gene. Compared with wild type Arabidopsis, overexpression of *SbMYBAS1* may reduce the salt tolerance of Arabidopsis. The yeast two-hybrid screening library results showed

that *SbMYBAS1* might interact with proteins encoded by SORBI_302G184600, SORBI_3009G247900, and SORBI_3004G59600. In addition, the expression of *AtGSTU17*, *AtGSTU16*, *AtP5CS2*, *AtUGT88A1*, *AtUGT85A2*, *AtOPR2*, and *AtPCR2* were regulated by *SbMYBAS1* to improve the salt sensitivity in Arabidopsis plants. This result is significant for exploring the salt response mechanism regulated by MYB TFs.

Materials and methods

Plant material

The salt-tolerant sweet sorghum inbred line M-81E was used as the experimental material. Select sorghum seeds of the same size, rinsed with tap water for 8 h, and then sowed. After emergence, the seedlings were irrigated with Hoagland nutrient solution every day. The germinated seeds were irrigated with Hoagland nutrient solution containing 5000 μM $\text{Ca}(\text{NO}_3)_2 \cdot 4\text{H}_2\text{O}$, 5000 μM KNO_3 , 2000 μM $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$, 1000 μM KH_2PO_4 , 46.25 μM H_3BO_3 , 9.14 μM $\text{MnCl}_2 \cdot \text{H}_2\text{O}$, 0.76 μM $\text{ZnSO}_4 \cdot \text{H}_2\text{O}$, 0.32 μM $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$, 0.11 μM $\text{H}_2\text{MoO}_4 \cdot \text{H}_2\text{O}$, 20.02 μM $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$, and 20.01 μM $\text{Na}_2\text{EDTA} \cdot 2\text{H}_2\text{O}$ (Wu et al. 2022). When it grew to 3 true leaves, it was treated with 150 mM NaCl for different hours (0, 6, 12, 24, 48, and 72), respectively, and then the materials were collected, and immediately stored at -80°C .

Identification of the MYB family in sorghum and analysis of the physicochemical properties of the proteins

The whole-genome sequence of *Sorghum bicolor* (NCBIv3.55) along with GFF3 (General feature format file) were downloaded from Ensembl Plants database (<https://www.plants.ensembl.org/>) (Yates et al. 2022). Additionally, Arabidopsis MYB amino acid sequences were retrieved from the Arabidopsis Information Resource (TAIR) (<http://www.arabidopsis.org/>) (Berardini et al. 2015) and the MYB protein sequence of Arabidopsis were used as the query sequence to search the MYB protein of sorghum. The MYB domain's HMM (Hidden Markov Model) profile was collected from the Pfam version 35.0 database (<https://www.interpro.entry/pfam/>) (Mistry et al. 2021). Following that, BLASTp ($E < 1e-5$) resemblance searches of MYB and variations of HMM builds ($E\text{-value} = 1e-3$) were being used to classify MYB in sorghum. The sequence was further analyzed by using SMART (<https://smart.embl.de/>) (Letunic et al. 2012) and PROSITE (<https://web.expasy.org/protpfam/>) (Duvaud et al. 2021).

Construction of phylogenetic tree of sorghum MYB family

A phylogenetic tree was constructed by combining the screened 210 sorghum MYB family members with 133 MYB genes in Arabidopsis, and a phylogenetic tree containing only members of the sorghum MYB family also was built. First, one step build a ML (maximum likelihood, bootstrap = 5000, JTT + I + G4) tree in TBtools was used to generate tree files (Chen et al. 2020), and then the ChiPlot website (<https://www.chiplot.online/>) for modification.

Analysis of conserved motif, domain and gene structure of sorghum MYB family

Conserved motifs in the full-length protein sequences of SbMYBs were predicted using the MEME suite (<https://meme-suite.org/meme/tools/meme>) (Nystrom and McKay 2021), then save the logo image of the Motif, and upload the MEME XML file in TBtools to draw the family member motif map (Chen et al. 2020). For the detection of conserved domains of each gene, we used the SMART website (<https://smart.embl.de/>) (Letunic et al. 2012). Upload the CDS sequence and full-length sequence of the gene on the GSDS 2.0 website (<http://gsds.gao-lab.org/>) (Moller-Leimkuhler and Muhleck 2020) to obtain the gene structure diagram, including CDS, upstream/downstream and intron.

Analysis of cis-acting elements of sorghum MYB family

The 2000 bp *cis*-acting element in the MYB family promoter region was obtained from the online website Plant CARE (<http://bioinformatics.psb.ugent.be/webtools/plantcare/>) (Rombauts et al. 1999) and after screening, the redundancy was deleted, *cis*-acting element diagram of the promoter was obtained by TBtools (Chen et al. 2020), and a total of six elements were reserved.

Analyses of chromosome distribution, gene duplication and synteny of MYB family in sorghum

The chromosome distribution information of *SbMYB* genes was obtained from the Ensembl Plant database (<http://plants.ensembl.org/>) (Yates et al. 2022). The gene duplication and synteny information was analyzed using TBtools (Chen et al. 2020). To estimate duplication events of *SbMYB* genes further, the K_a and K_s of evolution were calculated by using the Simple K_a/K_s Calculator. The divergence times (T) was

calculated as $T = Ks/2\lambda \times 10^{-6}$ Mya, the approximate value for the clock-like rate $\lambda = 6.5 \times 10^{-9}$ for sorghum.

Expression pattern analysis of sorghum MYB family gene under abiotic stress

The salt-tolerant sweet sorghum inbred line M-81E and the salt-sensitive inbred line Roma were treated with 150 mM NaCl for 24 h, and the samples were collected for transcriptome sequencing. By analyzing the transcriptome data, the gene expression heat map was drawn. The value of gene expression was averaged by the expression of three biological repeats. The differentially expressed genes were selected for analysis. M-81E is a salt-tolerant inbred line, and Roman is a salt-sensitive inbred line, they may have different ways to deal with salt stress, so some genes may have the opposite expression trend in salt treatment.

Generation of ectopic expression of Arabidopsis

SbMYBAS1 was ligated into the 1300–2 vector using Kpn I and BamH I restriction sites (Fig. S1) and transferred into *Agrobacterium tumefaciens* GV3101. The ectopic expression of *SbMYBAS1* in Arabidopsis was generated by inflorescence acquisition. The Arabidopsis seeds were harvested for selection in the ½ MS medium containing hygromycin and verified via RT-PCR analysis. The primers used in the experiments are listed in Table S1.

RNA extraction and qRT-PCR analysis: After RNA was extracted with Huayueyang RNA extraction kit (0416-50), cDNA was obtained by reverse transcription with Evo M-MLV reverse transcription kit (Code No. PCR analysis), confirm the amplification curve and melting curve of real-time quantitative PCR after the reaction, and use the $2^{-\Delta\Delta CT}$ method to analyze the data (Schmittgen and Livak 2008). Primers required for qRT-PCR (Table S1).

Subcellular localization and structure prediction of *SbMYBAS1* gene

Subcellular localization: (1) Transform the constructed 35S:*SbMYBAS1*-GFP vector and 35S: GFP empty vector into *Agrobacterium tumefaciens* GV3101. (2) Add the strain containing the target plasmid and the strain P19 with the helper plasmid to YEP medium containing relevant antibiotics overnight culture. (3) Activated bacterial solution: Add the bacterial solution from the second step to the new YEP medium, add acetosyringone at the same time, and shake the bacteria until OD600 = 0.9–1.0. (4) Discard the supernatant by centrifugation, and resuspend the bacteria with MMA body, until the OD value = 1.0 or so, and let it stand for 3 h in the dark. (5) Mix P19 with the target bacterial solution in equal proportions, inject tobacco with a syringe, and

cultivate normally for 36 h. (6) Tear off the lower epidermis of tobacco in the dark, and observe it under a laser scanning confocal microscope. SbMYBAS1 3D structure were predicted at the Swiss-model (<https://swissmodel.expasy.org/interactive>) online site (Waterhouse et al. 2018).

Physiological traits analysis of overexpression plants under salt stress

Wild-type (WT) and overexpressed Arabidopsis (OE) plants were treated with 100 mM NaCl and sampled after 7 days.

- (1) **DAB and NBT staining:** Add DAB and NBT staining solutions respectively, let stand in the dark for 8 h, then decolorize with 80% ethanol boiling water bath until the leaves fade, take pictures, and store with absolute ethanol (Jambunathan 2010).
- (2) **Determination of dry and fresh weight:** Pick plants with the same growth vigor, weigh their fresh weight, then put them in an oven at 65 °C, and weigh their dry weight one week later.
- (3) **Determination of total chlorophyll content:** Take 0.2 g of fresh leaves, extract with absolute ethanol, measure the absorbance values at wavelengths of 663 nm and 645 nm, and calculate the total chlorophyll content according to its formula (Hu et al. 2014).
- (4) **Determination of membrane permeability:** Weigh a certain amount of sample into a test tube, add ultrapure water, use a vacuum pump to create a vacuum environment, repeat 3 times, let stand for ion extravasation, and measure the conductivity value of the solution with a conductivity meter. This is the original conductivity value. After the sample is killed by high temperature, it is measured again. This is the total point conductivity value. According to the formula relative conductivity (%) = original conductivity value / total conductivity value * 100%, the permeability of plant cell membrane can be calculated.
- (5) **Determination of MDA content:** Determination of MDA content by thiobarbituric acid method (Ju et al. 2018).
- (6) **Determination of Na⁺, K⁺ content:** the material was boiled in water bath for 3 h, the volume was adjusted to 10 ml, filtered, and the ion content was measured with a flame spectrophotometer.

Data were subjected to analysis of variance and the means were compared using student's t-test at the 5% significance level using SPSS version 25.0 (IBM, Armonk, New York, NY, USA). The results were presented as mean ± standard error of the mean (n = 3), with significance values as follows: *, p < 0.05; **, p < 0.01; ***, p < 0.001; ****, p < 0.0001.

Self-activation activity verification and yeast two-hybrid screening library

Self-activation activity verification: construct *SbMYBAS1*-pGBKT7 vector, transform yeast AH109 competent cells, pick single colonies and dilute them in SD-Trp, SD-Trp/X- α -gal and SD-Trp/X- α -gal/AbA cultured on the medium.

Yeast two-hybrid screening library: transfer the *SbMYBAS1*-pGBKT7 plasmid into Y2H Gold yeast strain, and invert at 30 °C until the clone size is 2–3 mm; Pick a single clone and cultivate it in YPDA liquid medium; Add yeast library plasmid, BD plasmid and pre-denatured DNA and mix well; incubate it at 30 °C for 90 min; Resuspend with NaCl (0.9%), spread the bacterial solution on SD/Leu-Trp/X- α -Gal/AbA plate and cultivate at 30°C until the single clone grows and the primary screening is completed; Transfer the blue positive clones on the primary screening plate to the secondary screening medium D/Ade-His-Leu-Trp/X- α -Gal/AbA for further screening; culture at 30 °C until blue positive clones grow out one-to-one verification.

Results

Identification and physicochemical properties of MYB family members in sorghum

We searched the sorghum database in HMMER using the Pfam number (PF00249) of the MYB TF family. After deleting the repeated sequences, 210 sequences were obtained. The physicochemical properties of all protein sequences were predicted using ExPASy (Table S2). MYB family members were found to have around 78–1562 amino acids, 8.99–169.16 kDa relative molecular mass, and 4.48–10.68 isoelectric point. In the MYB protein family, 111 MYB proteins were found to be acidic, 3 neutral, and 96 neutral or alkaline.

Phylogenetic analysis of sorghum MYB family genes

To understand the taxonomic and evolutionary relationships of 210 MYB TFs in sorghum, a phylogenetic tree was constructed with 133 *A. thaliana* genes. These 343 genes were categorized into 6 taxa based on their sequence similarity with *A. thaliana* and the structure of the phylogenetic tree (Fig. 1). The subgroup I with the lowest number of genes only had 13 MYB genes, while the subgroup II with the highest number of genes had 116 MYB genes. In addition, the group III contained only 2 *A. thaliana* genes, while most other groups contained more *A. thaliana* genes. Most members of the II and IV groups belonged to the R2R3 MYB genes. This illustrates the changes incurred in the MYB family during the evolutionary process.

Motif, domain and structure analysis of MYB family members in sorghum

Based on the evolutionary analysis of these sequences, we divided these sequences into six groups (Fig. S2A). The conserved motifs in each group were similar, but the conserved motifs among different groups varied remarkably. The protein sequences containing motif 11 and motif 12 were closely related, which indicated that they had produced characteristic sequences different from other MYB proteins in the process of evolution. In addition, we observed that motif 3 was present in almost all MYB family members, indicating that motif 3 plays a vital role in influencing the function of the MYB family members. To explain the gene structure of the MYB family, we analyzed the introns and exons of MYB genes (Fig. S2B). In the sorghum MYB gene family, most members of the R2R3 MYB subfamily were composed of three exons and two introns. The numbers of exons and introns of other subfamilies were quite different, which indicated that R2R3 MYB subfamily genes may tend to form conservative functions in the process of evolution, while members of other subfamilies may have more functional differentiation in evolution. We uploaded the protein sequence of sorghum MYB gene to MEME website and displayed the analysis results (Fig. S2C). Twelve different motifs were distributed in different protein sequences, and the closely related genes have similar motif types and numbers. Different motifs also have different kinds of conservative amino acids, which was of great significance for the differentiation of protein function (Fig. S3).

A total of 210 MYB proteins from sorghum were checked against the SMART database to validate that they contained the MYB DNA-binding domain. At least one MYB DNA-binding domain was found in all MYB genes that encode proteins (Fig. S2D and Table S3). According to the number of DNA-binding domains, they were divided into four categories (1R, R2R3, 3R, 4R), and the conserved motifs of the DNA-binding domain sequences were analyzed by MEME, then found 5 conserved motifs. These domains had different characteristic amino acids, but arginine and tryptophan were more conservative (Fig. S4).

Analysis of cis-acting elements in the promoter region and protein function of MYB in sorghum

The *cis*-acting elements upstream of the MYB were predicted using the PlantCARE website. From a variety of *cis*-regulatory element, the keywords “response element” and “responsiveness element” were used to screen six corresponding types of elements, including various “motifs” and “boxes” (Table S4). Out of these 6 response elements, the majority were MeJA and ABA responsive elements. The *SbMYB62* and *SbMYB130* genes contained the highest number of ABA response

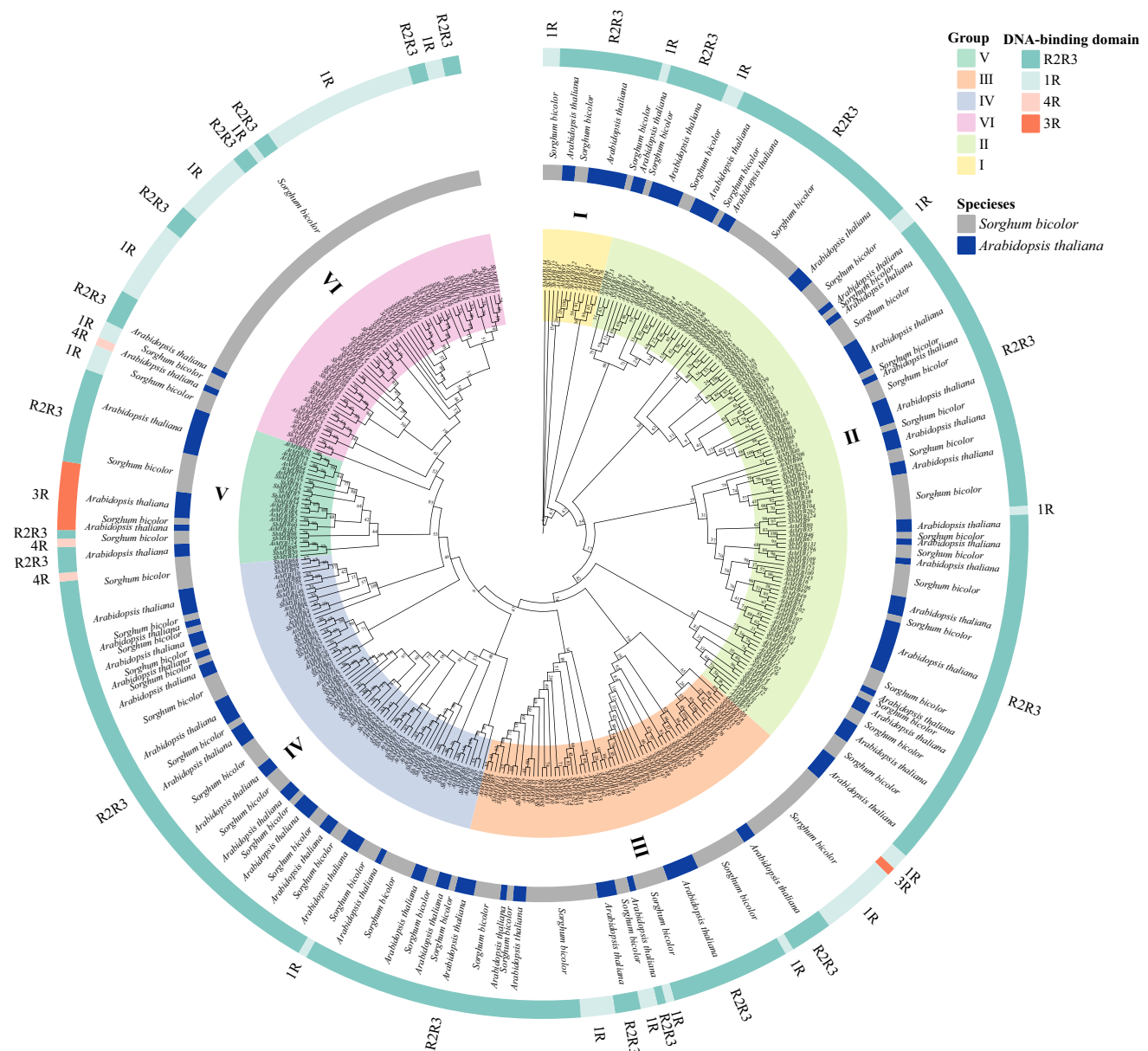


Fig. 1 Phylogenetic analysis of the MYB family in sorghum and *A. thaliana*. Using the protein amino acid sequences of sorghum and *A. thaliana* to construct a phylogenetic tree (maximum likelihood,

bootstrap=1000). They were divided into six (I–VI) subgroups, and *SbMYB* genes were divided into four different DNA-binding domain types

elements, and the *SbMYB189* gene contained the highest number of MeJA response element (Fig. 2A). Studies had shown that ABA and MeJA helped to improve the tolerance of plants to salt stress and other abiotic stresses. This showed that these three genes may be important to plant stress resistance.

Chromosomal location, gene duplication and synteny of *SbMYB* genes

Chromosomal localization analysis of 210 genes was performed, which showed that these genes were distributed on

10 chromosomes (chr1–10), as shown in (Fig. S5). However, the number and density of MYB genes on each chromosome were unevenly distributed. For instance, the lowest number of genes (8 genes) were found on chromosome 5, while chromosome 13 contained the highest number of genes (42 genes). In addition, MYB genes were mainly distributed in the upper and lower ends of chromosomes but not in centromeric regions. These genes were named *SbMYB1*–*SbMYB210* based on their position on the chromosomes (Table S2). There were 15 tandem duplication gene clusters distributed on seven chromosomes, the highest number

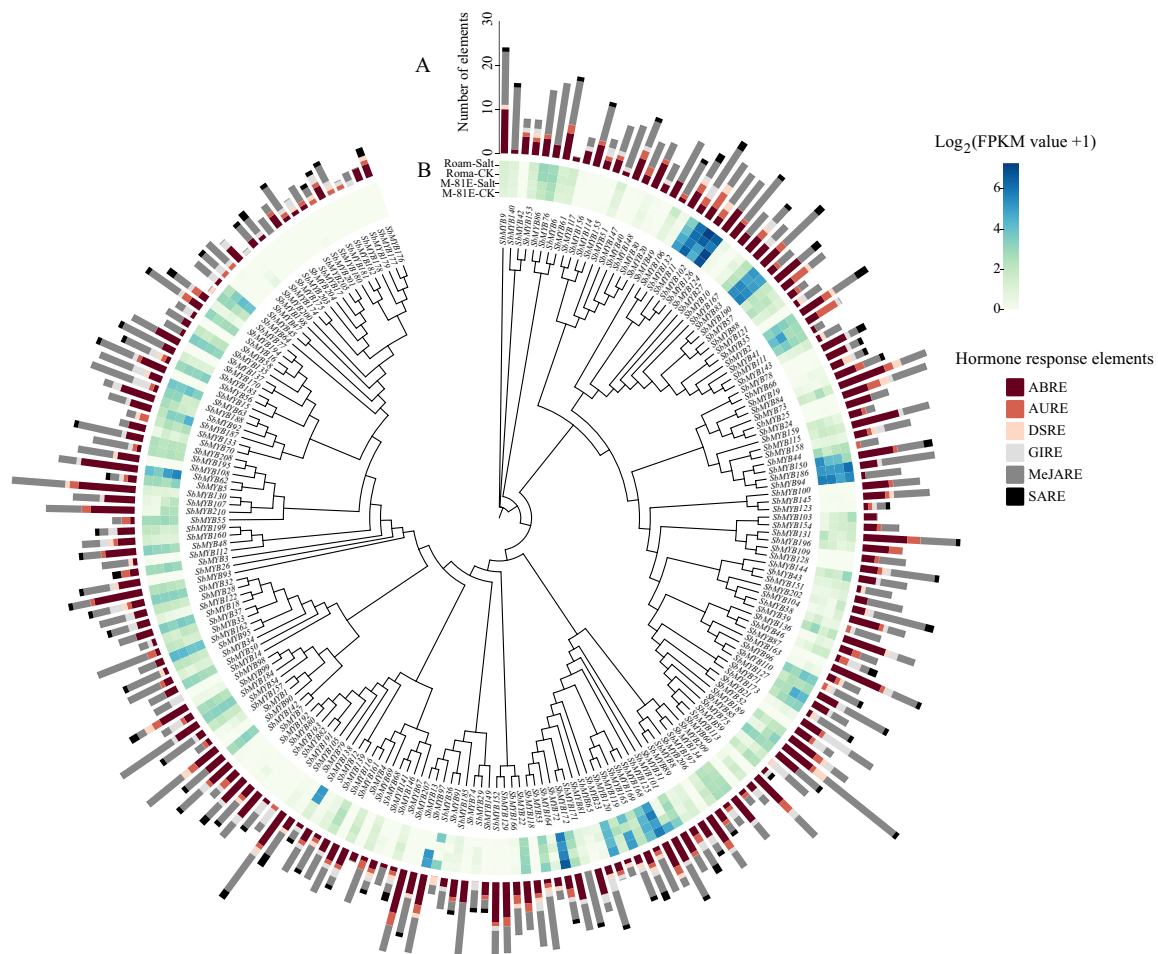


Fig. 2 Cis-acting elements and expression pattern of sorghum MYB family. **A** Type and number of response elements, **B** heat map of SbMYB genes expression in different sorghum cultivars under salt stress

of gene clusters were also found on chromosome 5. These genes may be important for the growth and development of sorghum.

Furthermore, to investigate the gene duplication events in sorghum, tandem duplications and segmental duplications were identified. Among the *SbMYB* genes, 35 tandem duplicated genes were identified and formed 20 pairs (Table S5). Meanwhile, 39 gene pairs generated from chromosomal segmental duplications were found (Fig. S6). To explore further the evolutionary process of *SbMYB* genes, genome synteny among *Zea mays*, *Setaria italica* and *Oryza sativa* was also investigated (Fig. 3A–C) and the results showed that 145, 141 and 124 *SbMYB* genes had homologous genes in *Zea mays*, *Setaria italica* and *Oryza sativa* (Table S5), respectively.

In addition, the substitution rate (non-synonymous/synonymous, Ka/Ks) was an effective index to determine the positive selection pressure after duplication and was typically used to understand the direction of evolution and its selective strength in a coding sequence (Lynch and Conery

2000). Ka/Ks = 1 stands for neutral selection, Ka/Ks < 1 means purifying selection, while Ka/Ks > 1 signifies positive selection (Lynch and Conery 2000). Therefore, we calculated the Ka, Ks and Ka/Ks of each gene pair. The average value of Ka for tandem duplication between *SbMYB* genes was significantly higher than the average for segmental duplication genes, but the opposite was found in Ks (Fig. 3D, E). Therefore, the Ka/Ks of tandem duplication genes is much higher than that of segmental duplication genes (Fig. 3F). The results showed that the Ka/Ks values of gene pairs of sorghum and gene pairs of *Zea mays*, *Setaria italica* and *Oryza sativa* orthologs were almost all < 1. The duplication events of tandem and segmental duplication genes were evaluated to have occurred approximately 46 and 64 Mya (million years ago). The divergence time was about 29.59053, 36.78345 and 49.08291 Mya for *Zea mays*, *Setaria italica* and *Oryza sativa*. These results implied that the duplication events played a significant role in evolution and functional divergence of *SbMYB* genes, as well as other Gramineae species. Further analysis showed that

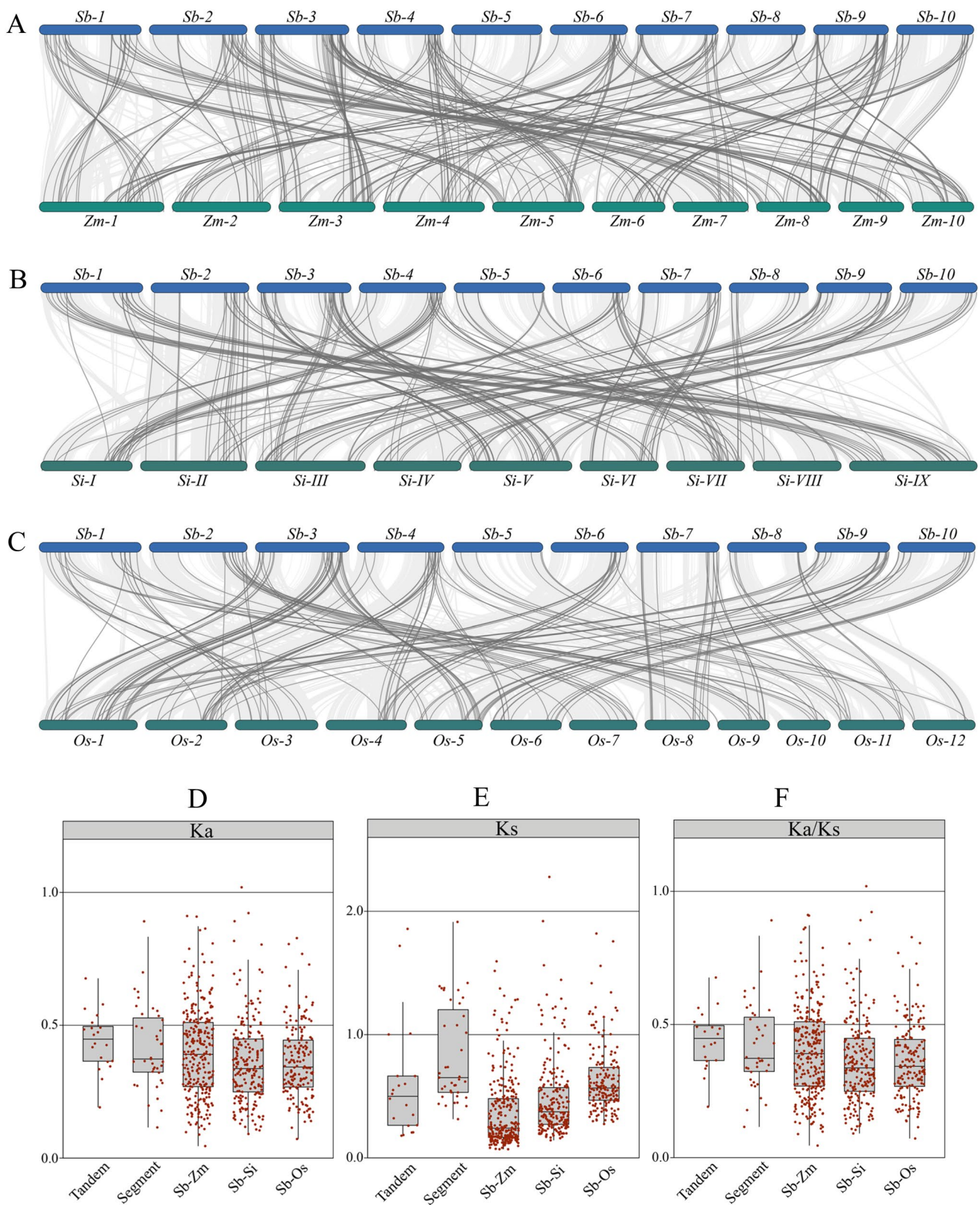


Fig. 3 Genes duplication and synteny of SbMYB genes. Comparative physical mapping shows the orthologous relationships of SbMYB genes with **A** *Zea mays*, **B** *Setaria italica* and **C** *Oryza sativa*. Gene pairs linked with blue lines. The horizontal axes in **D–F** stand for tan-

dem duplication (tandem), segmental duplication (segment) and the duplication between *Sorghum bicolor* and *Zea mays* (Sb-Zm), *Setaria italica* (Sb-Si) and *Oryza sativa* (Sb-Os)

SbMYB41 on chromosome 2 of sorghum had orthologous genes on chromosome 2 of four different Gramineous species (Fig. S7), and the promoter region of this gene had more auxin response elements than other *SbMYB* genes (Fig. 2A), so it may play a key role in the growth and development of sorghum.

Expression pattern analysis of sorghum MYB family members under salt stress

According to the transcriptome data (FPKM value), the expression levels of different genes under salt stress were analyzed (Table S6). For instance, the *SbMYB150* and *SbMYB187* were downregulated, while *SbMYB103* and *SbMYB164* was upregulated under salt stress conditions (Fig. 2B). Furthermore, the expression level of the same gene in different sorghum cultivars varied remarkably. For instance, *SbMYB172* was downregulated in the M-81E variety, while it was upregulated in the Roma variety of sorghum. Later, 11 differentially genes with apparent differences in expression levels were analyzed (Fig. S8). In the Roma variety of sorghum, the expression levels of *SbMYB39*, *SbMYB72*, *SbMYB168*, *SbMYB172*, *SbMYB186*, and *SbMYB189* were upregulated, whereas the expression level of *SbMYB169* was downregulated. In M-81E variety of sorghum, *SbMYB119*, *SbMYB150* and *SbMYB172* were downregulated, and *SbMYB41*, *SbMYB115*, and *SbMYB169* were upregulated. Also, the expression levels of MYB

family members changed significantly under salt stress conditions, which might have played a crucial role in regulating the response of plant to salt stress. Due to the maturity of gene editing technology and the non-introduction of foreign genes, we hoped to find a gene that negatively regulates salt tolerance in sorghum and carried out stable gene knock-out genetic transformation of sorghum. The expression of *SbMYB119* in Roma was higher than that in M-81E, and the gene expression was down-regulated under salt stress, so it was selected for follow-up study. *SbMYB119* was renamed as *SbMYBAS1* according to its annotation in NCBI.

Subcellular localization and expression of *SbMYBAS1*

SbMYB119 was renamed as *SbMYBAS1* according to its annotation in NCBI. The domain in *SbMYBAS1* was predicted, and We predicted the 3D structure of the protein (Fig. 4A, B). 35S: *SbMYBAS1*-GFP-transformed GV3101 *Agrobacterium* was injected into tobacco, and the location of *SbMYBAS1* was observed using a confocal laser microscope. The outcomes of the analysis showed a strong GFP fluorescent signal in the nucleus, while a single 35S: GFP signal appeared in the whole cell and nucleus. It can be seen that the *SbMYBAS1* was localized in the nucleus (Fig. 4C). In addition, the M-81E and Roma sorghum variety were treated with 150 mM NaCl for 0–72 h, and the expression levels of *SbMYBAS1* were analyzed. Under salt stress conditions,

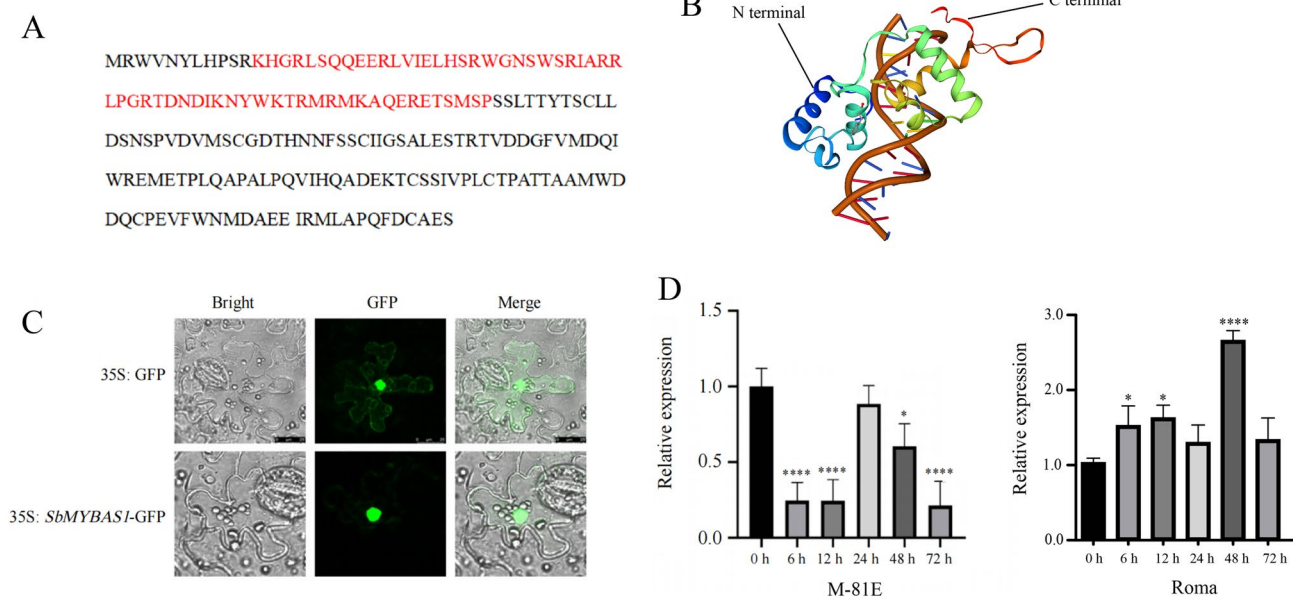


Fig. 4 Subcellular localization analysis, domain prediction, and expression analysis of *SbMYBAS1* gene. **A** Protein sequence of the *SbMYBAS1* gene. **B** MYB functional domain of the protein was predicted using the SMART website. **C** Subcellular localization of *SbMYBAS1* protein. **D** Expression level of *SbMYBAS1* under salt

stress in different sorghum varieties: qRT-PCR expression analysis of *SbMYBAS1* after NaCl treatment of sorghum for 0, 6, 12, 24, 48, and 72 h. Data represent mean values from three biological replicates ($\pm$ S.D.), *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$

SbMYBAS1 was downregulated, and its lowest expression was observed at 72 h in M-81E. SbMYBAS1 was upregulated, and its highest expression was observed at 48 h in Roma. This indicated that the expression level of *SbMYBAS1* decreased under salt stress (Fig. 4D).

Analysis of physiological traits of Arabidopsis plants overexpressing *SbMYBAS1*

Wild-type (WT) and *SbMYBAS1* over-expressed (OE) *A. thaliana* were treated with 100 mM NaCl, the phenotypes of the OE plants were more severely injured, and the leaves were wilted (Fig. 5A). The physiological properties of WT and OE were also determined. Since plants release a large amount of H₂O₂ and reactive oxygen species (ROS) under salt stress conditions, we investigated the levels of H₂O₂ and ROS synthesized in plants using DAB and NBT staining methods (Fig. 5B). The outcomes of the analysis showed that the OE plants produced more H₂O₂ and ROS post-salt treatment. Compared with WT, the dry and fresh weight of the OE plants was significantly lower after salt treatment (Fig. 5C, D). Under salt-stress conditions, the MDA level in OE plants was significantly higher than the WT (Fig. 5E). The germination rate of WT and the OE plants did not vary significantly under salt stress conditions (Fig. 5F). After salt treatment, the membrane permeability of both WT and OE plants increased, and it was higher (OE) than that of WT (Fig. 5G). Under salt stress conditions, the chlorophyll

content (WT) was higher than that of OE plants (Fig. 5H). Since salt stress leads to ion stress, we investigated the levels of Na⁺ and K⁺. Under salt stress conditions, Na⁺ content increased and K⁺ content decreased in plants. Compared to ion content of WT and OE plants before and after salt stress, we observed that Na⁺ level did not vary significantly, but the K⁺ level significantly decreased in OE plants. To compare the altered levels of ions more intuitively, we determined Na⁺ and K⁺ levels and found that the Na⁺/K⁺ value (OE) was higher than that of WT after salt treatment (Fig. 5I–K).

Validation of *SbMYBAS1* self-activation activity and interacting genes

We detected the salt-stress-related genes and analyzed their expression levels as shown in (Fig. 6A). In the verification experiment with Arabidopsis, the control was wild-type Arabidopsis and the experimental group was Arabidopsis strain with overexpression of *SbMYB119*. *AtGSTU18*, *AtGSTU17*, *AtGSTU16*, and *AtPRX52* are involved in abiotic stress (Chen et al. 2012b; Fernandez-Perez et al. 2015; Zhang et al. 2020a). The genes primarily involved in osmotic regulation are *AtP5CS2*, *AtUGT88A1*, *AtUGT85A2*, *AtGLH*, and *AtPLA2A*. *AtPCR2* plays a vital role in plant response to ion stress. *AtUBL* is involved in the ubiquitination pathway. The expression levels of *AtGSTU17*, *AtGSTU16*, *AtP5CS2*, *AtUGT88A1*, *AtUGT85A2*, *AtOPR2*, and *AtPCR2* were

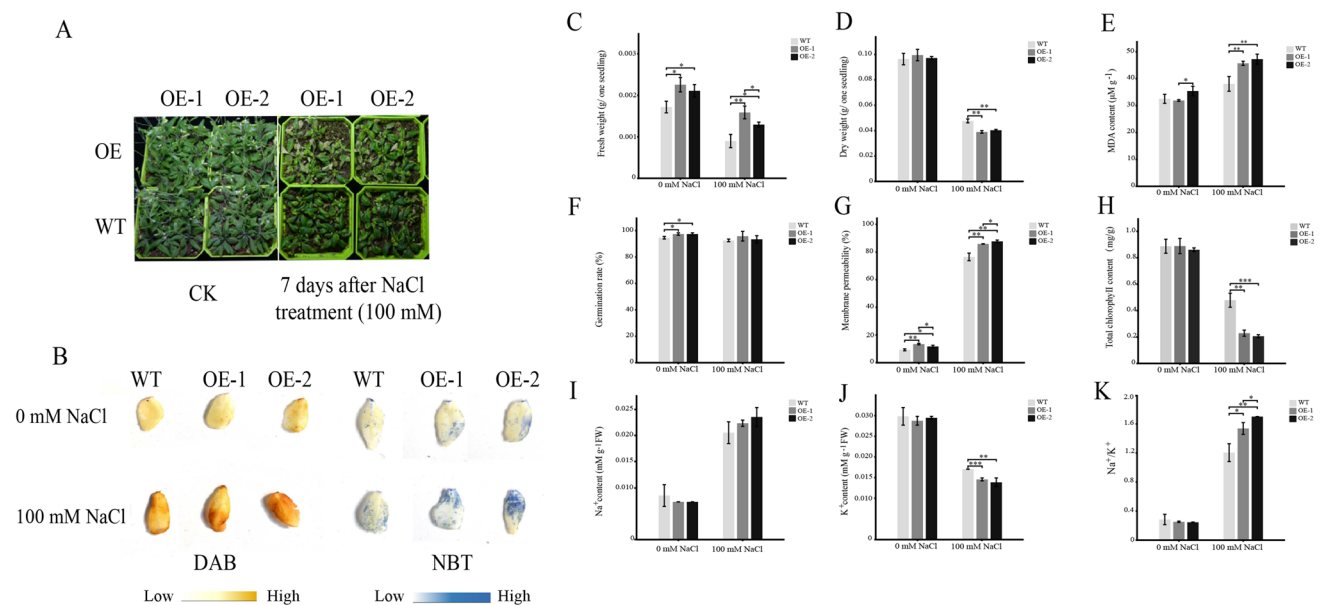


Fig. 5 Physiological traits of Arabidopsis plants overexpressing *SbMYBAS1* treated with salt stress. **A** Phenotypes of WT and transgenic plants after one week of 100 mM NaCl treatment. **B** Dry and fresh weight, germination rate, chlorophyll content, membrane permeability, MDA content, and Na⁺ and K⁺ contents in transgenic

plants and WT, respectively. **C** Leaves of WT and transgenic plants were treated with 0 and 100 mM NaCl and stained with DAB and NBT staining solution, respectively. Data represent mean values from three biological replicates ($\pm$ S.D.), *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$

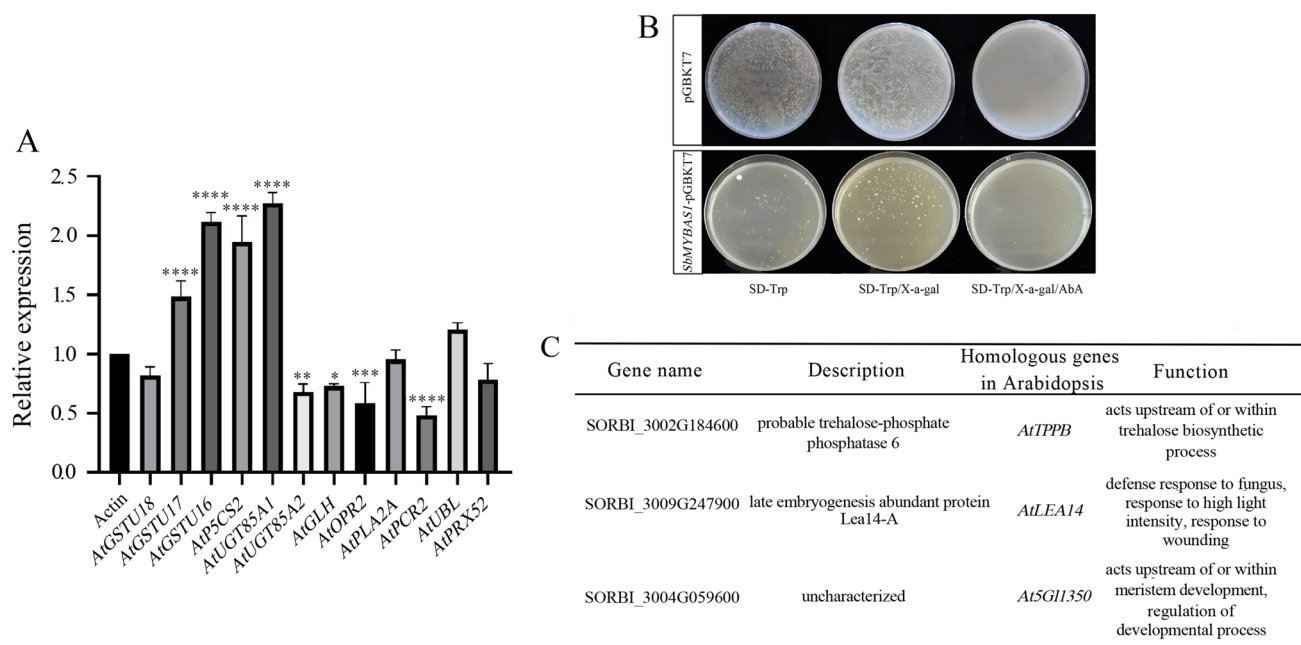


Fig. 6 Validation of SbMYBAS1 self-activation activity, screening of interacting genes, and expression analysis of salt-related genes. **A** The relative expression levels of salt-related genes were analyzed by qRT-PCR. Data represent mean values from three biological replicates ($\pm$ S.D.), *, $p < 0.05$; **, $p < 0.01$; ***, $p < 0.001$; ****, $p < 0.0001$.

significantly altered. *AtGSTU17* and *AtGSTU16* genes encode glutathione transferase enzyme, which is involved in transporting non-enzymatic antioxidant glutathione in reactive oxygen scavenging systems. *AtP5CS2* encodes a rate-limiting enzyme involved in the proline biosynthesis, which improves plant's tolerance to stress by regulating the osmotic process. *AtUGT88A1* and *AtUGT85A2* genes encode glycosyltransferases that participate in osmotic regulation by regulating the synthesis of soluble sugars. *AtOPR2* is involved in the biosynthesis of jasmonic acid. Jasmonic acid and its derivatives are endogenous hormones which affect salt resistance of plants. *AtPCR2* is a gene associated with plant tolerance to cadmium ions, which significantly induced cadmium stress resistance in Arabidopsis plants (Reddy et al. 2023).

We transformed pGBKT7 or pGBKT7-*SbMYBAS1* plasmid into competent yeast cells. The colonies transformed yeast cells did not turn blue in SD-trp/X- α -gal/AbA medium, indicating that *SbMYBAS1* did not possess self-activating activity (Fig. 6B). Later, we screened the interaction proteins through the yeast screening library. Thus, we screened three potential interacting proteins. As shown in Fig. 6C, trehalose-6-phosphate phosphatase, late embryogenesis Abundant protein, and the function of one protein is unknown.

B After transforming the pGBKT7 or pGBKT7-*SbMYBAS1* plasmid into competent yeast, the transformed yeast cells were spread into the medium containing SD-trp, SD-trp/X- α -gal, and SD-trp/X- α -gal/AbA. **C** Yeast two-hybrid based screening of *SbMYBAS1* interacting genes

Discussion

MYB TF plays a crucial role in controlling plant growth and development, primary/secondary metabolism, and the response of plant to biotic and abiotic stresses (Tan et al. 2020). In this research, we identified 210 MYB genes, which were divided into seven taxa. Isoelectric points are correlated to the pH value and salt concentration of its environment (Shi et al. 2005). Analysis of the physical and chemical properties of these genes showed that the isoelectric point of 69 genes was less than seven. We observed that motif 1 was present in almost all MYB family members. This might explain the importance of this motif. In addition, the motifs vary significantly among different groups. This showed that although the functions of most motifs are unknown, genes in the same group might have similar functions due to similar motifs (Feller et al. 2011).

The MYB gene consists of 2–3 exons, 1–2 two introns, and sequences at the upstream and downstream ends. Members of the same group have an identical number of introns/exons, positions, and degrees. Homology of gene structure and conserved motifs indicate the closeness of the evolutionary relationship and functional similarity. Different structures of the MYB family member indicate

MYB family members with new functions, suggesting functional diversity in the MYB family members (Yang et al. 2019).

Analysis of gene expression patterns under salt stress led to the isolation of differentially regulated genes in sorghum, and *SbMYBAS1* was selected for further analysis.

When plants are subjected to a salt stress condition, the permeability of the plasma membrane increases, and the inner membrane systems of various organelles expand, shrink, or get damaged. This is primarily due to membrane lipid peroxidation, membrane protein denaturation, altered membrane lipid fluidity, phase transition, and membrane structure damage. Plants stimulate ROS production under salt stress conditions, which are the primary culprit for the peroxidation of membrane lipids (Kong et al. 2016). The degree of membrane lipid peroxidation of plant cell membranes is determined by measuring the MDA level, as the MDA level is directly related to the degree of stress.

In this study, MDA content (OE) was significantly higher than that of WT. This indicated that membrane lipid peroxidation level was higher in plants overexpressing *SbMYBAS1*. K^+ plays a crucial role in plant growth and development, while Na^+ is not required in most plants. When extracellular Na^+ concentration is high, a high amount of Na^+ enters the cytoplasm, which interferes with the K^+ transport, leading to ion and nutritional stresses. The equilibrium of Na^+ and K^+ transport is achieved at high Na^+ (Hmidi et al. 2019). In this study, Na^+ content and the Na^+/K^+ ratio increased in OE plants. This suggests that *SbMYBAS1* may affect salt tolerance of plants by regulating ion transport.

We performed yeast two-hybrid screening and screened three potential *SbMYBAS1* interacting genes (SORBI_3002G184600, SORBI_3009G247900, and SORBI_3004G59600). SORBI_3002G184600 encodes trehalose-phosphate phosphatase 6, and its homologous gene in Arabidopsis is *AtTPPB*, which mainly encodes trehalose-6-phosphate phosphatase and participates in the trehalose pathway. Trehalose is an essential signaling metabolite that is related to sugar signal transduction. Current research on *AtTPPB* showed that it regulates plant growth and development through the trehalose pathway and enhance plant's tolerance to low temperature (Nunes et al. 2013; Ponnu et al. 2011). Enhanced trehalose metabolism could improve the plant's tolerance to abiotic stress. Therefore, the *SbMYBAS1* gene may be involved in the sugar signal transduction pathway by interacting with the *SbMYBAS1* protein to regulate the plants' tolerance to salt stress conditions. SORBI_3009G247900 encodes the late embryogenesis abundant protein Lea14-A, and its ortholog in Arabidopsis is *AtLEA14*, which encodes a late embryogenesis abundant protein (LEA), a cellular stress-related protein. LEA protein is mainly expressed in late embryonic development, and its expression and accumulation indicate seed maturity

(Singh et al. 2005). LEA protein imparts protective effects against drying, oxidative stress, and osmotic stress. *AtLEA* protein was detected only when amaranth was subjected to salt stress. This suggests that LEA protein may also play an important role in regulating salt stress (Saucedo et al. 2017).

As demonstrated in a previous study, overexpression of the *SiLEA14* improved the salt and drought tolerance in millets (Wang et al. 2014). Ectopic expression of the *AdLEA* in tobacco increased the plant's tolerance to dehydration, oxidative stress, and salt stress (Sharma et al. 2016), which further indicates that LEA protein enhances plant salt tolerance. *SbMYBAS1* gene regulates salt resistance of plants by regulating LEA protein synthesis. SORBI_3004G059600 is uncharacterized, and its Arabidopsis homolog *At5G21050* encodes DNase I-like superfamily protein. Few studies have explored this protein and have suggested that this protein might be related to the regulation of inorganic phosphate in root response to salt stress (Kawa et al. 2016).

Quantitative analysis of salt-related genes demonstrated significant upregulation or downregulation of several genes. Both *AtGSTU16* and *AtGSTU17* genes encode glutathione transferase, which plays a crucial role in abiotic stress in plants. In Arabidopsis, a total of 55 glutathione transferases have been reported so far. *AtGSTU16* and *AtGSTU17* belong to the Tau class (Dixon and Edwards 2010). It was found that *AtGSTU17* plays a negative regulatory role in drought and salt stress signaling pathways, and its Arabidopsis mutant *AtGSTU17* increases the resistance of plants to drought and salt stress conditions (Chen et al. 2012a). *AtGSTU16* is also associated with lead (Pb) tolerance (Zhang et al. 2020b). *AtP5CS2* encodes delta1-pyrroline-5-carboxylate synthase, which affects proline biosynthesis under abiotic stress conditions and can catalyze the first step of proline synthesis from glutamate. Proline plays a crucial role in osmotic regulation by protecting enzyme activity and maintaining redox balance. Previous studies on *AtP5CS2* have shown that *AtP5CS2* was located in the cytoplasm of the cell and is necessary for the root growth of seedlings under abiotic stress conditions. Therefore, the loss of the *AtP5CS2* gene increases plants tolerance to salt stress conditions (Funk et al. 2020). *AtUGT88A1* and *AtUGT85A2* encode UDP-glucosyltransferase (UGT), which regulates a variety of metabolites (such as plant hormones) and enzymatic activities by interacting with glucuronidase (GUS), affecting plant development (Woo et al. 2007).

AtOPR2 gene encodes 12-oxo-phytodienoic acid reductases (OPR). According to the substrate specificity, *OPR_s* can be divided into two groups, and *AtOPR2* belongs to group I. However, it was found that *AtOPR2* does not participate in the jasmonic acid signaling pathway. When plants are injured, the expression level of *AtOPR2* is increased, but its exact function remains unclear (Dong et al. 2013; Li et al. 2009). *AtPCR2* mainly encodes zinc

transport and detoxification membrane proteins, which plays a crucial role in maintaining ion homeostasis (Song et al. 2010). Further studies using various techniques, such as monoclonal, EMSA, ChIP, LUC, and so on (Tang et al. 2021), should explore if *SbMYBAS1* regulates these genes.

Conclusion

A total of 210 MYB transcription factors were identified in sorghum. To analyze the expression patterns of these genes under salt stress, we finally selected *SbMYBAS1* for further study. Under salt stress conditions, transgenic *Arabidopsis* plants had lower dry and fresh weight and chlorophyll content and higher membrane permeability, MDA level, and Na^+/K^+ ratio than WT. Compared with WT, transgenic *Arabidopsis* plants suffered more damage under salt stress. *SbMYBAS1* overexpression decreased the tolerance of plants to salt stress. It might be due to the interaction of *SbMYBAS1* with the protein encoded by SORBI_3002G184600, SORBI_3009G247900, and SORBI_3004G059600 to affect the expression levels of downstream target genes, regulating salt stress tolerance of plants. This study is significant for further research on the MYB TF family in sorghum in response to salt stress conditions, which may improve salt tolerance and increase the yield of sorghum plants.

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Data availability The sorghum MYB protein sequences were collected from Ensembl Plants database (<https://www.plants.ensembl.org/>). The *Arabidopsis* MYB protein sequences were downloaded from TAIR database (<http://www.arabidopsis.org/>). RNA-Seq data were downloaded from NCBI SRA database under the following accession number, SRR11147325, SRR11147326, SRR11147327, SRR11147328, SRR11147329, SRR11147330, SRR11147331, SRR11147332, SRR11147333, SRR11147334, SRR11147335 and SRR11147336.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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